Discrete Burgers Circuit Dislocation Analysis (Method 2)

Theoretical Formulation and Implementation in the MTM Lattice-Triangulation Code

Technical Note

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Abstract

In 2D nonlinear finite-element and atomistic crystal plasticity simulations on Delaunay triangulations, topological vertex valence (`NodeValence` / `ReferenceCoordination`, counting triangles incident to each atom) works on six-fold triangular lattices ($p = 6$) where every elementary triangle is equilateral, but breaks down on four-fold square lattices ($p = 4$). Because each $a_0 \times a_0$ square unit cell is split by a diagonal of length $\sqrt{2}\,a_0$ chosen between two degenerate orientations (`/` and `\`), any boundary between `/` and `\` diagonal domains produces vertices of triangle valence 5 and 7 in a 100% defect-free crystal. This note presents the theoretical formulation and implementation of **Method 2 (Discrete Burgers Circuit)**, a discrete exterior calculus formulation of Nye's dislocation density tensor $\boldsymbol{\alpha} = -\operatorname{Curl}(\boldsymbol{F}_e^{-1})$. By projecting directed mesh edges onto the ideal Bravais lattice as an anti-symmetric $\mathbb{Z}^2$-valued 1-cochain $\Delta\boldsymbol{n}_{ab}$ and taking its discrete exterior derivative $\boldsymbol{B}_T = (d\Delta\boldsymbol{n})_T$ around each counter-clockwise triangle T, Method 2 guarantees $\boldsymbol{B}_T \equiv \boldsymbol{0}$ on every defect-free crystal configuration (regardless of `/` vs. `\` diagonal switches or integer surface steps), satisfies Discrete Stokes' Theorem identically across arbitrary patches and dislocation junctions, and localizes dislocation cores to single elementary triangles.

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1 Why Topological Triangle Valence Fails on Square Lattices

Let $\mathcal{T} = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ be a Delaunay triangulation of N atoms with deformed positions $\{\boldsymbol{y}_i\}_{i=1}^N \subset \mathbb{R}^2$. The topological triangle coordination number of vertex $i \in \mathcal{V}$ is the number of incident triangles:

$$z_i^{\text{tri}} = \#\{T \in \mathcal{F} : i \in T\}. \tag{1}$$

In a 2D triangular (hexagonal) Bravais lattice ($p = 6$) with lattice spacing a_0, each bulk atom has 6 nearest neighbors at distance a_0, every elementary triangle is equilateral with side lengths

(a_0, a_0, a_0), and $z_i^{\text{tri}} = 6$ everywhere in a defect-free crystal. An isolated edge dislocation corresponds to a 5/7 coordination dipole ($z_i^{\text{tri}} = 5$ adjacent to $z_j^{\text{tri}} = 7$).

In a 2D square Bravais lattice $\mathbb{Z}^2 a_0$ ($p = 4$), however, each atom has only **four** nearest neighbors at distance a_0 and **four** second-nearest neighbors at distance $\sqrt{2}\, a_0$. Every elementary $a_0 \times a_0$ square plaquette is cocircular (its four vertices lie on a circle of diameter $\sqrt{2}\, a_0$) and must be split into two right-angled isosceles triangles of side lengths $(a_0, a_0, \sqrt{2}\, a_0)$ by inserting either:

- the **positive diagonal** (/) connecting $(m, n) \leftrightarrow (m + 1, n + 1)$, or
- the **negative diagonal** (\) connecting $(m + 1, n) \leftrightarrow (m, n + 1)$.

At any interior atom i, the four surrounding square cells always contribute 4 axis-aligned nearest-neighbor bonds, plus $d_i \in \{0, 1, 2, 3, 4\}$ diagonal bonds of length $\sqrt{2}\, a_0$ that happen to touch atom i. Hence the triangle valence at atom i is:

$$z_i^{\text{tri}} = 4 + d_i \in \{4, 5, 6, 7, 8\}. \tag{2}$$

Whenever floating-point roundoff or minuscule elastic strain causes adjacent square cells to switch between / and \ diagonals, d_i takes values 1 and 3, producing $z_i^{\text{tri}} = 5$ and 7 in a completely defect-free square lattice.

2 Continuum vs. Discrete Multiplicative Plasticity ($\boldsymbol{F} = \boldsymbol{F}_e \boldsymbol{F}_p$)

2.1 Kröner–Bilby Decomposition and Nye's Dislocation Tensor

In the Mesoscopic Tensorial Model (MTM) on a fixed reference triangulation, the total element deformation gradient $\boldsymbol{F} = \nabla \boldsymbol{y}$ is decomposed multiplicatively as

$$\boldsymbol{F} = \boldsymbol{F}_e \boldsymbol{F}_p, \tag{3}$$

where $\boldsymbol{F}_p \in \mathrm{GL}(2, \mathbb{Z})$ is the lattice-invariant plastic slip obtained via Lagrange reduction $\boldsymbol{C}_{\text{red}} = \boldsymbol{M}^{\mathsf{T}} \boldsymbol{C} \boldsymbol{M}$ with $\boldsymbol{M} = \boldsymbol{F}_p^{-1} \in \mathrm{SL}(2, \mathbb{Z})$ ($\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}} \boldsymbol{F}$), and $\boldsymbol{F}_e = \boldsymbol{F} \boldsymbol{F}_p^{-1} = \boldsymbol{F} \boldsymbol{M}$ is the **elastic distortion tensor**.

Pulling back an infinitesimal deformed line element $d\boldsymbol{y}$ by $\boldsymbol{F}_e^{-1}$ yields the local elastic lattice differential:

$$d\boldsymbol{\xi} = \boldsymbol{F}_e^{-1}\, d\boldsymbol{y}. \tag{4}$$

In a compatible (defect-free) crystal, $d\boldsymbol{\xi}$ is an exact differential ($\mathrm{Curl}(\boldsymbol{F}_e^{-1}) = \boldsymbol{0}$). Around a dislocation core, $d\boldsymbol{\xi}$ is no longer closed, and its circulation around a counter-clockwise contour $\mathcal{C} = \partial\Omega$ defines the true continuum Burgers vector:

$$\boldsymbol{b}(\mathcal{C}) = -\oint_{\mathcal{C}} \boldsymbol{F}_e^{-1}\, d\boldsymbol{y} = -\int_{\Omega} \mathrm{Curl}(\boldsymbol{F}_e^{-1})\, dA, \tag{5}$$

where $\boldsymbol{\alpha} = -\,\mathrm{Curl}(\boldsymbol{F}_e^{-1})$ is **Nye's dislocation density tensor**.

2.2 Effect of Topological Remeshing on $\boldsymbol{F}_p$

When adaptive Delaunay remeshing (`AdaptiveMesher`) is enabled in the MTM code, topological edge flips reconnect atoms to their *new* physical neighbors after each plastic slip event and reset the element reference configuration. Consequently, in a region where a dislocation has already glided through and left a slipped perfect crystal behind, the remeshed element deformation gradients satisfy $\boldsymbol{F}_T \approx \boldsymbol{I}$ ($\boldsymbol{M}_T = \boldsymbol{I}$): the plastic slip $\boldsymbol{F}_p$ has been absorbed into the **graph connectivity** of the mesh.

Method 2 evaluates the discrete counterpart of Eq. (5) directly on the directed edges of the current triangulation $\mathcal{T}$, making it invariant to whether plastic slip was accommodated by $\boldsymbol{M}_T \in \mathrm{SL}(2, \mathbb{Z})$ or by Delaunay reconnection.

3 Discrete Exterior Calculus Formulation of Method 2

3.1 Reference Frame Pull-Back and Automatic Bravais Basis $\boldsymbol{A}_0$

For each vertex a of an active triangle $T \in \mathcal{F}$ with node index n_a and periodic translation vector $\boldsymbol{t}_a$, we first pull back the position by the macroscopic external deformation gradient $\boldsymbol{F}_{\text{ext}}$:

$$\boldsymbol{x}_a = \boldsymbol{F}_{\text{ext}}^{-1}\, \boldsymbol{y}_{n_a} + \boldsymbol{t}_a \in \mathbb{R}^2. \tag{6}$$

Each triangle $T = (0, 1, 2)$ is oriented counter-clockwise in the reference frame so that $(\boldsymbol{x}_1 - \boldsymbol{x}_0) \times (\boldsymbol{x}_2 - \boldsymbol{x}_0) > 0$.

The ideal Bravais lattice basis matrix $\boldsymbol{A}_0 \in \mathbb{R}^{2\times 2}$ of the crystal is defined as

$$\boldsymbol{A}_0 = a_0\, \boldsymbol{R}(\theta_0)\, \boldsymbol{B}_{\text{lat}}, \qquad \boldsymbol{R}(\theta_0) = \begin{pmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{pmatrix}, \tag{7}$$

where the primitive dimensionless basis $\boldsymbol{B}_{\text{lat}}$ is

$$\boldsymbol{B}_{\text{lat}}^{\text{square}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (p = 4), \qquad \boldsymbol{B}_{\text{lat}}^{\text{triangular}} = \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix} \quad (p = 6). \tag{8}$$

The parameters (p, a_0, θ_0) are extracted automatically from $\mathcal{T}$:

1. For each triangle T with sorted edge lengths $L_1(T) \le L_2(T) \le L_3(T)$, the median ratio $\text{median}_T(L_3/L_1)$ separates triangular lattices ($\approx 1.0 < 1.20$, $p = 6$) from square lattices ($\approx \sqrt{2} \approx 1.414 > 1.20$, $p = 4$).
2. The lattice constant a_0 is taken from `userData->ideal_lattice_parameter` (with fallback to $\text{median}_T(L_1)$).
3. The global crystal orientation angle $\theta_0 \in (-\pi/p, \pi/p]$ is computed from all nearest-neighbor edges ($L_{ab} \in [0.75\, a_0, 1.22\, a_0]$) via the p-fold bond-orientational order parameter:

$$Z_p = \sum_{(a,b)\in\mathcal{E}_{\text{NN}}} e^{ip\phi_{ab}}, \qquad \theta_0 = \frac{1}{p}\,\text{atan2}(\text{Im}\, Z_p, \text{Re}\, Z_p). \tag{9}$$

3.2 Canonical Anti-Symmetric Lattice Step 1-Cochain $\Delta\boldsymbol{n}_{ab}$

On every directed edge $e_{ab} = (a \to b) \in \mathcal{E}$, we map the relative vector $\boldsymbol{x}_b - \boldsymbol{x}_a$ into the ideal Bravais basis and round to the nearest integer lattice vector $\Delta\boldsymbol{n}_{ab} \in \mathbb{Z}^2$. To ensure that $\Delta\boldsymbol{n}$ is a strict **algebraic 1-cochain** ($\Delta\boldsymbol{n}_{ba} \equiv -\Delta\boldsymbol{n}_{ab}$ even when a component lies near a half-integer tie ± 0.5), we canonicalize the edge direction using the lexicographical order $(n_a, x_a, y_a) \prec (n_b, x_b, y_b)$:

$$\Delta\boldsymbol{n}_{ab} = \begin{cases} \text{round}\Big(\boldsymbol{A}_0^{-1}(\boldsymbol{x}_b - \boldsymbol{x}_a)\Big) & \text{if } a \prec b, \\ -\,\text{round}\Big(\boldsymbol{A}_0^{-1}(\boldsymbol{x}_a - \boldsymbol{x}_b)\Big) & \text{if } b \prec a. \end{cases} \tag{10}$$

3.3 Elementary Triangle Burgers Circuit (2-Cochain $\boldsymbol{B}_T$)

For each counter-clockwise triangle $T = (0, 1, 2) \in \mathcal{F}$ (excluding non-physical outer boundary slivers with $L_3(T) > 2.15\, a_0$), the discrete exterior derivative $d\Delta\boldsymbol{n}$ around ∂T defines the integer Burgers vector of triangle T:

$$\boldsymbol{B}_T = (d\Delta\boldsymbol{n})(T) = \Delta\boldsymbol{n}_{01} + \Delta\boldsymbol{n}_{12} + \Delta\boldsymbol{n}_{20} \in \mathbb{Z}^2, \tag{11}$$

and the dimensionless physical Burgers vector (in units of a_0):

$$\boldsymbol{b}_T = \frac{1}{a_0}\boldsymbol{A}_0\boldsymbol{B}_T \in \mathbb{R}^2, \qquad b_T = \|\boldsymbol{b}_T\|_2. \tag{12}$$

Proposition 1 (Exact Vanishing on Defect-Free Square and Triangular Lattices)**.** *Let $\Omega \subset \mathcal{T}$ be any defect-free crystal region in which each atom $a \in \Omega$ is displaced from an integer Bravais site $\boldsymbol{m}_a \in \mathbb{Z}^2$ by an elastic displacement $\boldsymbol{u}_a$ ($\boldsymbol{x}_a = \boldsymbol{A}_0\boldsymbol{m}_a + \boldsymbol{u}_a$) satisfying the elastic compatibility bound $\|\boldsymbol{A}_0^{-1}(\boldsymbol{u}_b - \boldsymbol{u}_a)\|_\infty < \frac{1}{2}$ on every edge (a, b) in Ω. Then $\boldsymbol{B}_T \equiv (0, 0)^\mathsf{T}$ for every triangle $T \subset \Omega$, regardless of whether square diagonals are oriented as / or \ or whether boundaries contain integer surface steps.*

Proof. Since $\|\boldsymbol{A}_0^{-1}(\boldsymbol{u}_b - \boldsymbol{u}_a)\|_\infty < \frac{1}{2}$, Eq. (10) evaluates to the exact lattice difference $\Delta\boldsymbol{n}_{ab} = \boldsymbol{m}_b - \boldsymbol{m}_a = (d\boldsymbol{m})_{ab}$. Because $\Delta\boldsymbol{n} = d\boldsymbol{m}$ is an exact 1-coboundary and $d^2 \equiv 0$, summing around any triangle $T = (0, 1, 2)$ yields

$$\boldsymbol{B}_T = (\boldsymbol{m}_1 - \boldsymbol{m}_0) + (\boldsymbol{m}_2 - \boldsymbol{m}_1) + (\boldsymbol{m}_0 - \boldsymbol{m}_2) \equiv \boldsymbol{0}. \qquad \square$$

3.4 Discrete Stokes' Theorem, Plaquette Cancellation, and Junctions

Proposition 2 (Discrete Stokes' Theorem and Frank's Rule)**.** *For any patch of triangles $\Omega = \bigcup_{k=1}^{M} T_k \subset \mathcal{F}$ with counter-clockwise outer boundary loop $\partial\Omega$, the circulation of $\Delta\boldsymbol{n}$ around $\partial\Omega$ equals the exact integer sum of elementary triangle Burgers vectors inside Ω:*

$$\boldsymbol{B}(\partial\Omega) \equiv \oint_{\partial\Omega} \Delta\boldsymbol{n}_{ab} = \sum_{T \subset \Omega} \boldsymbol{B}_T \in \mathbb{Z}^2. \tag{13}$$

Proof. Every interior edge shared by two triangles $T_1, T_2 \subset \Omega$ is traversed once as $a \to b$ and once as $b \to a$. By Eq. (10), $\Delta\boldsymbol{n}_{ab} + \Delta\boldsymbol{n}_{ba} \equiv \boldsymbol{0}$ in exact integer arithmetic, leaving only the outer boundary edges $\partial\Omega$. $\square$

Two key consequences follow directly from Proposition 2:

1. **Unit-Cell Plaquette Cancellation Across Shared Square Diagonals:** In a square lattice, two triangles T_1, T_2 sharing a diagonal edge of length $L_{\text{shared}} \approx \sqrt{2}\,a_0 > 1.15\,a_0$ form a single 4-atom unit-cell plaquette $Q = T_1 \cup T_2$ containing no interior atoms. If only the shared internal diagonal edge crosses a rounding threshold under elastic shear while the four outer unit-cell bonds remain compatible, then $\boldsymbol{B}(Q) = \boldsymbol{B}_{T_1} + \boldsymbol{B}_{T_2} = \boldsymbol{0}$ with $\boldsymbol{B}_{T_1} = -\boldsymbol{B}_{T_2} \neq \boldsymbol{0}$. Cancelling such opposite pairs ($\boldsymbol{B}_{T_1} = \boldsymbol{B}_{T_2} = \boldsymbol{0}$) removes single-diagonal elastic rounding artifacts while leaving $\sum_{T \subset \Omega} \boldsymbol{B}_T$ invariant on every larger circuit.

2. **Nodal 1-Ring Circuits and Dislocation Junctions:** Summing $\boldsymbol{B}_T$ over the 1-ring $\Omega_i = \bigcup_{T \ni i} T$ of each atom i yields the nodal 1-ring Burgers vector:

$$\boldsymbol{B}_i^{(\text{1-ring})} = \sum_{T \ni i} \boldsymbol{B}_T \in \mathbb{Z}^2, \qquad \boldsymbol{b}_i^{(\text{1-ring})} = \frac{1}{a_0}\boldsymbol{A}_0\boldsymbol{B}_i^{(\text{1-ring})}. \tag{14}$$

   When two dislocations with Burgers vectors $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ intersect or react to form a junction (e.g. $\boldsymbol{B}_1 = (1, 0)^\mathsf{T}$ and $\boldsymbol{B}_2 = (0, 1)^\mathsf{T}$ in `Shift / Staircase`), Eq. (13) automatically enforces Frank's conservation rule $\boldsymbol{B}_{\text{junction}} = \boldsymbol{B}_1 + \boldsymbol{B}_2 = (1, 1)^\mathsf{T}$.

4 Summary of Exported ParaView VTK Fields

Table 1 lists the VTK fields written by `ConfigurationSaver::writeToVTKFile` and `ConfigurationSaver::writeToVTK_DefectAnalysis`.

Table 1: Method 2 Discrete Burgers Circuit fields exported to binary VTK files.

VTK Field Name	**Section**	**Type**	**Physical Meaning**
`ElementBurgersMagnitude`	`CELL_DATA`	`float`	$\|\boldsymbol{b}_T\|_2$ per triangle in units of a_0 (0 in perfect crystal, 1 at elementary dislocation core)
`ElementBurgersVector`	`CELL_DATA`	`float[3]`	$(b_{T,x}, b_{T,y}, 0)$ per triangle in units of a_0
`BurgersMagnitude`	`POINT_DATA`	`float`	$\|\boldsymbol{b}_i^{(\text{1-ring})}\|_2$ around the 1-ring of atom i
`BurgersVector`	`POINT_DATA`	`float[3]`	$(b_{i,x}^{(\text{1-ring})}, b_{i,y}^{(\text{1-ring})}, 0)$ around the 1-ring of atom i
`ReferenceCoordination`	`POINT_DATA`	`float`	Burgers-filtered coordination (6 whenever $\boldsymbol{B}_i^{(\text{1-ring})} = \mathbf{0}$; 5/7 only at $\boldsymbol{B}_i^{(\text{1-ring})} \neq \mathbf{0}$ cores)
`NodeValence`	`POINT_DATA`	`float`	Unfiltered mesh triangle valence z_i^{tri}